

BAMM PECHS GOVT COLLEGE FOR WOMEN

Preliminary Examination, 2026

Mathematics PAPER – II

(Pre-Engineering & computer science Group)

Max.Marks: 100

SECTION "A"

(MULTIPLE CHOICE QUESTIONS-M.C.Q's)

Marks: 20

Q1. Choose the correct answer for each from the given options:

- i. If $y = \tan^{-1}\sqrt{x}$, then $\frac{dy}{dx}$ is:
a) $\frac{1}{1+x^2}$ b) $\frac{1}{\sqrt{x+x}}$ c. $\frac{1}{2(x\sqrt{x+x})}$ d. Not defined
- ii. $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-7n}$:
a) e b) 1 c) e^{-7} d) ∞
- iii. In maple to reset all variables we use the command
a) >Restart b) >reset c) >clear d) >cancel
- iv. If the line coincides with x-axis then its equation is:
a) $y = b$ b) $-b$ c) $y = 0$ d) ∞
- v. $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ is equal to:
a) e b) -1 c) 1 d) 0
- vi. Is the function $\vec{F}(t) = \cos t\vec{i} + \sin 2t\vec{k}$ at $t = \pi$:
a) is not continuous b) is continuous c) limit doesn't exist d) None of these
- vii. The point (3, 3) is ___ the circle $x^2 + y^2 = 64$.
a) inside b) outside c) on d) can't determine
- viii. $\int_2^2 (x^3 + 3x^2 - 5x^{\frac{-1}{2}}) dx$, is:
a) 0 b) $12 - 10\sqrt{2}$ c) $24 - 20\sqrt{2}$ d) $20\sqrt{2}$
- ix. If a function $f(x)$ is such that $f'(c) = 0$, then the point $(c, f(c))$ is called"
a) Maximum point b) Stationary point c) Minimum point d) Critical point
- x. Auxiliary circle of ellipse $\frac{x^2}{6} + \frac{y^2}{5} = 1$:
a) $x^2 + y^2 = 36$ b) $x^2 + y^2 = 25$ c) $x^2 + y^2 = 5$ d) $x^2 + y^2 = 6$
- xi. $\int f^n(x) f'(x) dx$, where $n = -1$, is:
a) $\ln|f^n(x)| + c$ b) $\frac{f^{n-1}(x)}{n-1} + c$ c) $\frac{f^{n+1}(x)}{n+1} + c$ d) $\ln|f(x)| + c$
- xii. If the eccentricity is zero than the conic is?
a) Circle b) Ellipse c) Hyperbola d) Parabola
- xiii. The degree of differential equation $y = x \left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx}$ is:

- a) 0 ☒ b) 2 c) 1 d) 3
- xiv. What is the length of tangent drawn from an external point $(-1, 2)$ to the circle $2x^2 + 2y^2 - 4x + 8y =$
 a) $\sqrt{30}$ b) 15 c) 30 d) $\sqrt{15}$
- xv. The set $\{x|x \in \mathbb{R}\}$ can be written as:
 a) $(-\infty, \infty)$ b) $[-\infty, 1)$ ☒ c) $[-\infty, \infty]$ d) $[\infty, -\infty)$
- xvi. Which function is transcendental function:
 a) Exponential b) Hyperbolic c) Inverse Trigonometric ☒ d) All of them
- xvii. Two lines are said to be parallel if and only if their slopes are:
☒ a) Equal b) Unequal c) Negative reciprocal of each other d) Undefined
- xviii. The perimeter of rectangle is given by a function $P(x, y) = 2(x + y)$, where x and y are length and breadth. The sum of partial derivative w.r.t their independent variables is:
 a) $2x$ ☒ b) $2y$ c) $2(x + y)$ d) 4
- xix. $\frac{d}{dx}(e^n) =$
 a) $n e^{n+1}$ ☒ b) $n e^{n-1}$ c) $(n + 1) e^{n+1}$ d) 0
- xx. The degree of differential equation $y = x \left(\frac{dy}{dx} \right)^2 + \frac{dy}{dx}$ is:
 a) 0 ☒ b) 2 c) 1 d) 3

SECTION "B" (SHORT ANSWER QUESTIONS)

Marks: 40

Attempt any TEN Questions from this section. All questions carry equal marks.

- Find by definition (ab-initio) the derivatives w.r.t x of: $f(x) = (6x + 7)^{\frac{5}{2}}$ OR $f(x) = \sin 2x$
- ii. If $f(x) = \frac{x+1}{2}$ and $g(x) = 2x - 1$ are two given bijective functions then prove $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- OR Find the limit of the sequence $1, \frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \dots$
- iii. Find the limit of the following (any 1): a) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\cos \sec^2 x - 2}{\cot x - 1}$ b) $\lim_{x \rightarrow \pi} \frac{\sin(\pi - x)}{\pi(\pi - x)}$
- iv. Evaluate by using integration by parts: $\int \cos(\ln x) dx$ OR $\int e^x \cdot \sin x dx$
- v. Find the limit of the following (any 1): a) $\lim_{x \rightarrow 0} \frac{\sin x^0}{x}$ b) $\lim_{h \rightarrow 0} \frac{(1+2h)^n - 1}{5h}$
- vi. If $f(x) = f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$ Find the value of k if the given ftn is continuous.
- vii. Show that: $\tan \{3 \tan^{-1}(\frac{x}{3})\}$ Prove that $y' = 4 \left(\frac{1+y^2}{4+x^2} \right)$. OR $\cot H = 5 = \sec = 1.10$
- If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots}}}$, shw that: $(2y - 1)y' = \cos x$.
- viii. If $A(2, 5)$, $B(3, 7)$ and $C(0, 8)$ are vertices of a triangle then find the equation of altitude through B .
OR Find the equation of line which passes through $(-2, -4)$ and the sum of its intercept equals to 3.
- ix. Find the derivative by using chain rule: $y = (x + \sqrt{x^2 + 1})^n$ OR Find $\frac{dy}{dx}$ when $y = x^x$
- x. Find equation of circle passing through two points $(1,0)$ and $(0,1)$ and touch the line $x+y=0$.
- xi. Find the equation of parabola whose vertex is $(1,2)$ and directrix is $y=4$.
- xii. Find the equation of hyperbola with Centre at origin and transverse axis is along y -axis such that latusrectum is 12 unit and eccentricity is 2.
- xiii. Obtain differential equation by eliminating constant, $y = A \sin 2x + B \cos 2x$
- xiv. Evaluate the integral $\int x \cos^{-1} x dx$ $\frac{x^2}{2} - \frac{x^2}{2} - \frac{x^4}{4} - \frac{1}{4}$
- xv. The gradient of one of the lines of the lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other,
Show that $8h^2 = 9ab$. OR Prove that the tangent lines to the curve $y^2 = 4ax$ at points where $x = a$ are at right angles to each other.

SECTION "C" (DETAILED ANSWERS QUESTIONS)

Marks: 40

NOTE: Answer FIVE questions from this section. Each carry 8 marks.

3. Evaluate the following integral (any TWO)
 $\int 2x^3 e^x dx$ $\int \tan^3 x \sec^3 x dx$ $\int x \ln x dx$
4. A rectangular reservoir with a square bottom and open top is to be lined inside with lead. Find the dimension of the reservoir to hold $\frac{1}{2} a^3$ cubic meters, such that the lead required is minimum. OR
 Find the relative maximum and minimum values of the function $f(x) = x^4 - 8x^2$ OR $x^2 - 3x + 1$.
5. Given that $u = \sec^{-1} \left(\frac{x^2 - y^2}{x + y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$. OR
 Given that $u = \sin^{-1} \left(\frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$

6. Obtain McLaurin series for $\sin(x)$. OR Find the Taylor series of function $\ln(1+x)$ near $a=1$.
7. The vertices A, B and C of a triangle are (2, 1), (5, 2) and (3, 4) respectively Find the coordinates of the circumcenter and also the radius of the circum-circle of the triangle. OR Find the equation of a line through the intersection of the lines $x + 2y - 5 = 0$ and $3x - 2y + 1 = 0$ and perpendicular to the line $2x - 3y + 7 = 0$.
8. Find centre, foci, vertices and latusrectum of the conic $9x^2 - 18x + 4y^2 + 16y - 11 = 0$. Also draw its graph.
9. Find eq of tangent and normal to $x^2 + y^2 = 100$ at (6,8).
10. A ball is thrown upward vertically with velocity 49m/s. find a) the time when the body at maximum height, b) find the height with $t=3$ sec, c) find maximum height.